



CURT  
CARIOCA  
DOPPIO  
MACCHIATO  
ESPECIARIAS

CAFÉ FILTRADO  
PRENSA FRANCESA

IRISH COFFEE

② Macchiato

LATTE MACCHIATO  
MOCHA

③ Cappuccinos

CAPPUCCINO ITALIANO  
CAPPUCCINO CREMA  
CAPPUCCINO FIT

④ Lattes

LATTE  
SWISS CHOCOLATE

# Order and supply management system for a café

Specialty Coffee & Lifestyle Café

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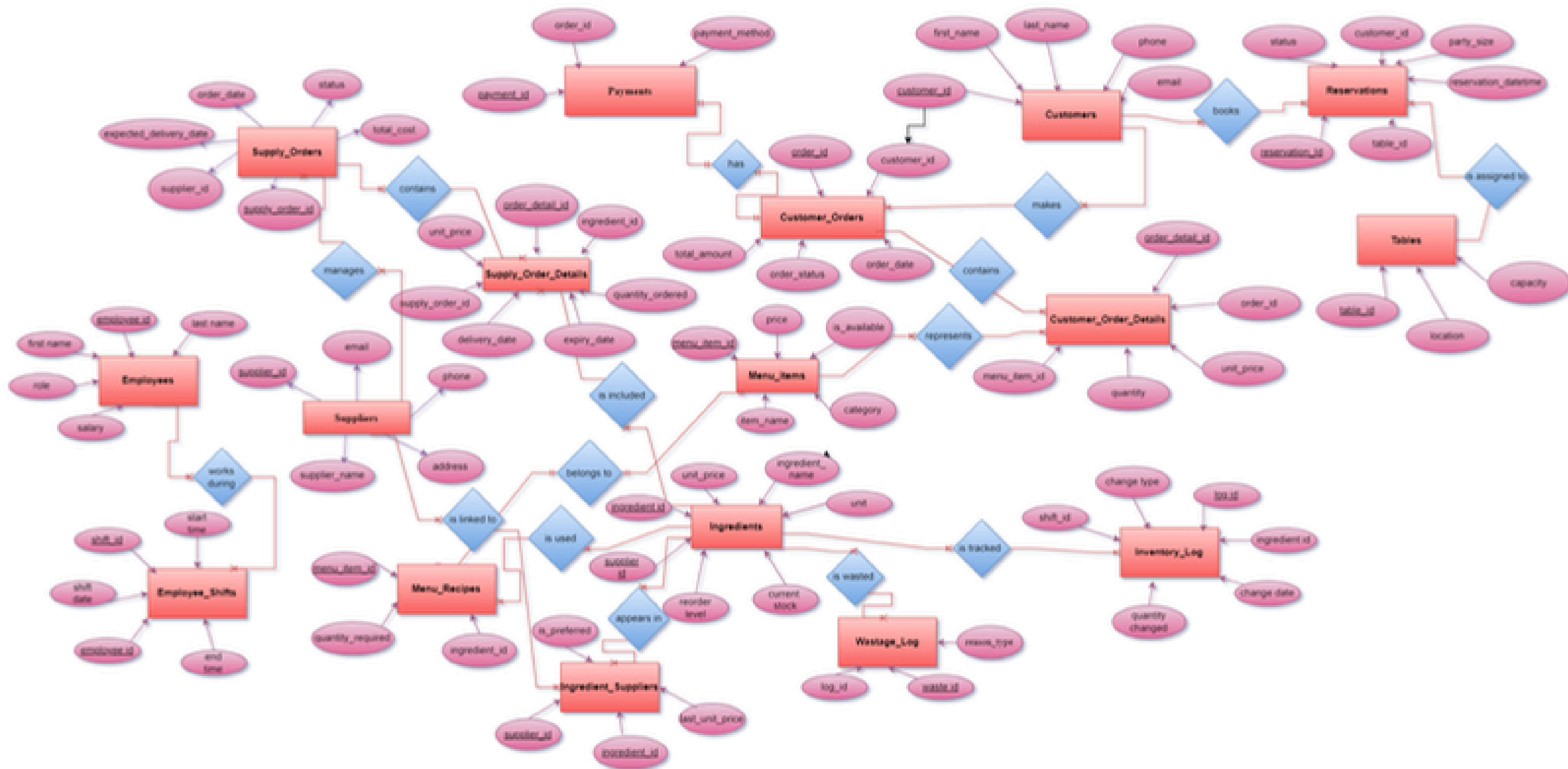


**Running a café is not only about cooking good food – it is a continuous balance between inventory, suppliers, menu performance, customers, staff productivity, and financial control.**



Our Order & Supply Management System (OSMS) centralizes all these processes into a single, well-structured data model and provides powerful analytics that help the café operate efficiently every day.







# Inventory & Supply Management Analysis



PROCEDURE pr\_supplier\_performance

Shows id, name, n of orders, n of deliveries, days and total cost



TRIGGER trg\_inventory\_log\_update\_stock



TRIGGER trg\_order\_served\_usage

TRIGGER trg\_supply\_delivered\_purchase



TRIGGER trg\_expiry\_to\_waste



## SUPPLIER PERFORMANCE

|               |                     |          |        |                     |                    |
|---------------|---------------------|----------|--------|---------------------|--------------------|
| Supplier ID=1 | Name=Alonzo Pogue   | Orders=2 | Late=1 | Avg delay=5 days    | Total cost=1250.73 |
| Supplier ID=2 | Name=Darci Schimoni | Orders=1 | Late=0 | Avg delay=0 days    | Total cost=926.73  |
| Supplier ID=3 | Name=Melina Gowen   | Orders=4 | Late=3 | Avg delay=6.67 days | Total cost=1818.94 |
| Supplier ID=4 | Name=Dinny Mumford  | Orders=5 | Late=1 | Avg delay=0 days    | Total cost=2430.33 |

## **TRIGGER** `trg_update_menu_availability`

We use this trigger whenever the stock level of an ingredient changes, specifically when `current_stock` becomes 0 or negative, meaning the ingredient is no longer available for cooking.

## **PACKAGE** `pkg_menu_insights`

### **PROCEDURE** `pr_hero_items`

This procedure identifies the top-selling items within each menu category, based on revenue from Ready (completed) customer orders. It prints the highest performers — your café's "Hero Items" — using both quantity sold and total revenue.

### **PROCEDURE** `pr_dead_items`

This procedure finds "dead" or underperforming menu items — items that were ordered very rarely. It scans all menu items and reports those that are not generating sales.





# PROCEDURE pr\_top\_selling\_items



## CURSOR

TYPE t\_sales\_row IS RECORD



TYPE t\_sales\_tab IS TABLE OF t\_sales\_row;

Why we use it:

- Helps identify the most profitable menu items.
- Shows which categories perform best (coffee, desserts, main dishes, snacks).
- Helps plan inventory and supply orders based on real demand.
- Supports menu optimization and marketing decisions.
- Useful for dashboards, management reports, and business strategy.



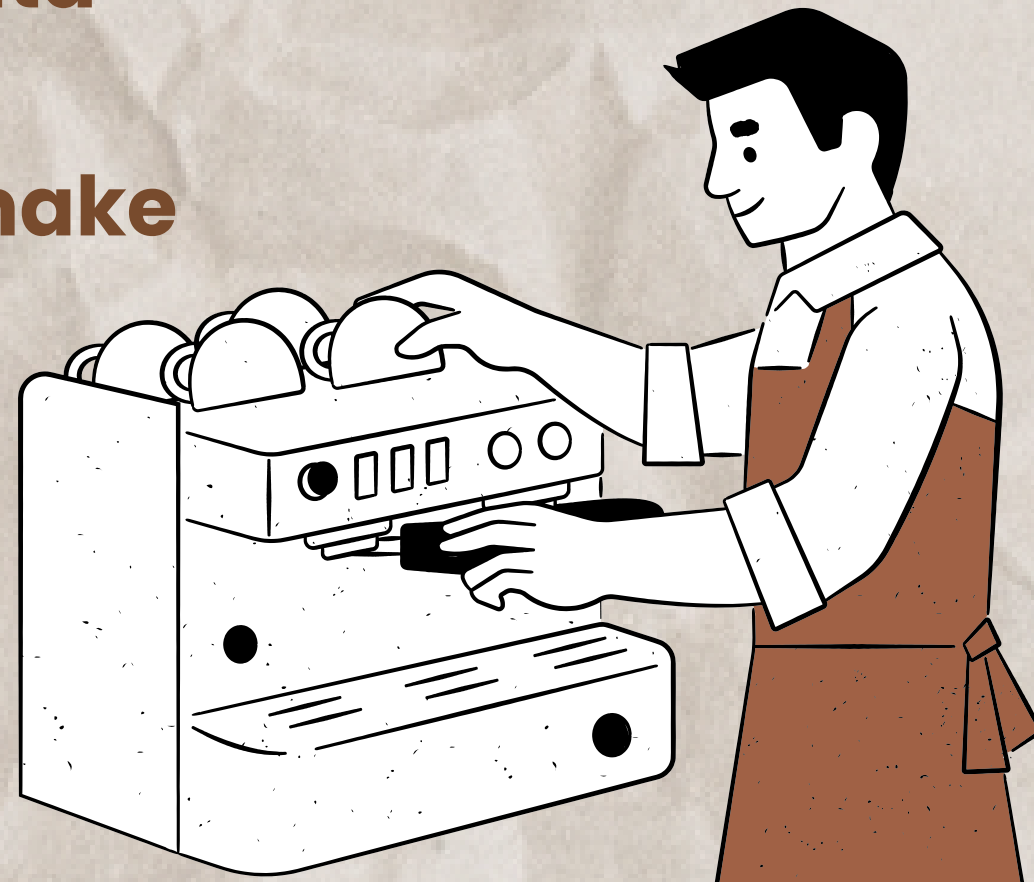
# PROCEDURE

## pr\_menu\_profit\_report

☕ By using cursor

### Purpose

- Analyzes how profitable each menu item is in a given period.
- Combines sales data (what customers ordered) with cost data (how much ingredients cost).
- Helps understand which items make money and which items are less profitable.



# Customer behavior analysis(pkg\_customer\_behavior)

- PROCEDURE pr\_customer\_order\_summary
- FUNCTION fn\_customer\_favorite\_item
- PROCEDURE pr\_customer\_recommendation\_report
- TYPE t\_category\_nt IS TABLE OF VARCHAR2(100);



## What it does

- **Analyzes each customer's ordering habits.**
- **Provides easy-to-understand customer statistics.**

## Main features

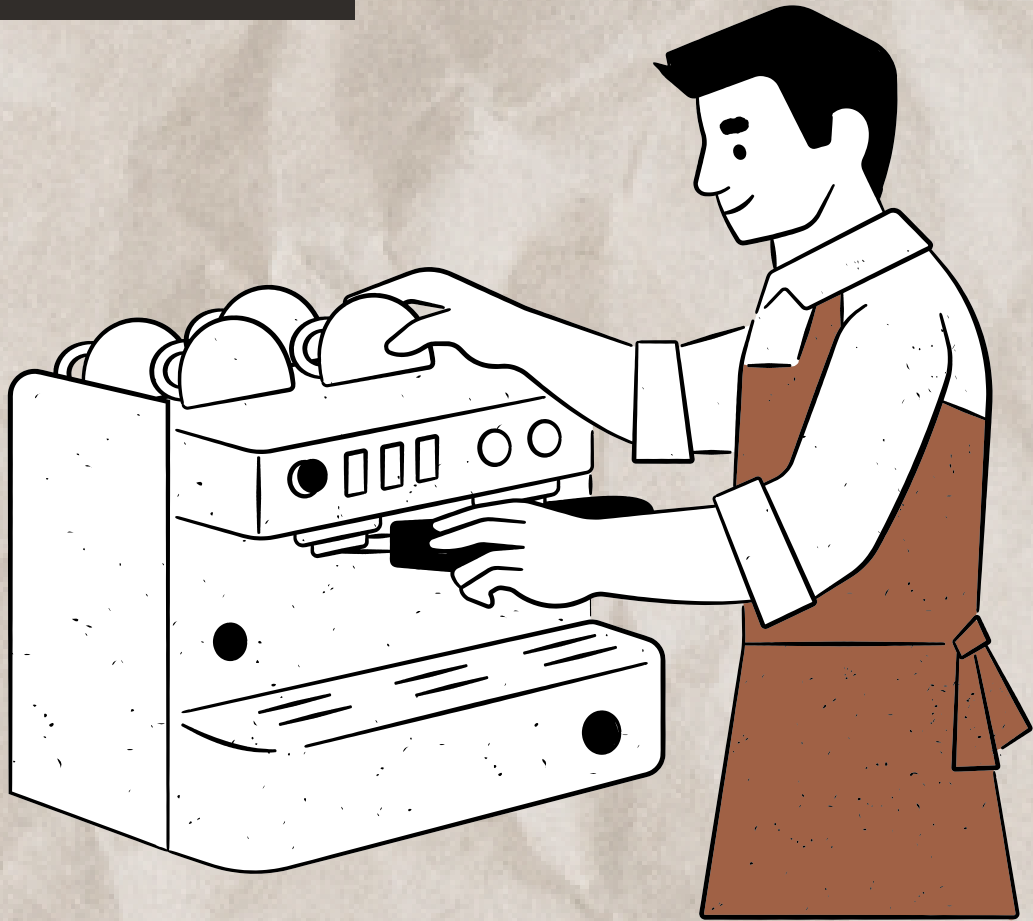
- **Order summary: total orders, total money spent, first and last order dates.**
- **Favorite item: finds the customer's most frequently ordered menu item.**
- **Recommendations: identifies the customer's favorite category (coffee, desserts, etc.) and their top 3 favorite items.**

# Outputs

| Results                                                                                                                                                                                             | Explain | Describe | Save |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|------|
| === Customer Order Summary ===<br>Customer: Olivie Belli (ID: 66)<br>Total orders: 10<br>Total spent: 67<br>First order date: 2025-11-12<br>Last order date: 2025-12-07<br><br>Statement processed. |         |          |      |

| Results                                                   | Explain | Describe | Save |
|-----------------------------------------------------------|---------|----------|------|
| Favorite item: Peach Iced Tea<br><br>Statement processed. |         |          |      |

|                                                                                                                                                                                                 |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| === Customer Recommendation Report ===<br>Customer ID: 66<br>Top category: Dinner (total items ordered: 7)<br>Top 3 recommended items:<br>1) Peach Iced Tea<br>2) Chicken Wrap<br>3) Fried Rice |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|



## Why it's useful

- Helps understand customer preferences.
- Supports personalized offers and promotions.
- Increases customer loyalty and boosts revenue.



# Reservation and tables analysis(pkg\_reservation\_and\_tables)



FUNCTION fn\_reservation\_showup\_rate



PROCEDURE

pr\_reservation\_showup\_statistics



PROCEDURE

pr\_table\_performance\_summary

- Calculates the overall reservation show-up rate for a given period.
- Builds daily show-up statistics (total reservations, show-ups, no-shows, % rate)
- Summarizes table performance: reservations per table and average party size, which table has the most reservations

## Main goals:

- *Understand how reliable reservations are.*
- *Detect no-show problems and patterns.*
- *Analyze table usage and capacity efficiency.*



# Output

```
=== FN: Show-up rate ===  
Rate: 75.93%
```

```
=== PR: Daily show-up stats ===
```

```
=== Reservation Show-Up Statistics ===
```

```
Period: 2025-11-10 .. 2025-11-20
```

```
2025-11-10 | total=0, showups=0, no_shows=0, rate=0%  
2025-11-11 | total=12, showups=9, no_shows=3, rate=75%  
2025-11-12 | total=10, showups=8, no_shows=2, rate=80%  
2025-11-13 | total=18, showups=13, no_shows=5, rate=72.22%  
2025-11-14 | total=12, showups=11, no_shows=1, rate=91.67%  
2025-11-15 | total=11, showups=7, no_shows=4, rate=63.64%  
2025-11-16 | total=8, showups=4, no_shows=4, rate=50%  
2025-11-17 | total=9, showups=8, no_shows=1, rate=88.89%  
2025-11-18 | total=13, showups=10, no_shows=3, rate=76.92%  
2025-11-19 | total=6, showups=5, no_shows=1, rate=83.33%  
2025-11-20 | total=9, showups=7, no_shows=2, rate=77.78%
```

```
=== PR: Table performance summary ===
```

```
=== Table Performance Summary ===
```

```
Period: 2025-11-10 .. 2025-11-20
```

```
Table 15 (Terrace, capacity=4) | reservations=7, avg_party=2  
Table 7 (Corner, capacity=4) | reservations=7, avg_party=2  
Table 18 (Counter, capacity=8) | reservations=6, avg_party=5  
Table 5 (Window, capacity=6) | reservations=6, avg_party=4  
Table 19 (Center, capacity=3) | reservations=6, avg_party=2  
Table 4 (Terrace, capacity=2) | reservations=5, avg_party=1  
Table 16 (Counter, capacity=3) | reservations=5, avg_party=2  
Table 24 (Counter, capacity=8) | reservations=5, avg_party=4
```

I also do a trigger  
`trg_reservation_capacity_check`

```
Reservation OK: party_size=4 fits in table 1 (capacity=6)  
  
1 row(s) inserted.
```

```
Our cafe follows the laws of physics – choose a larger table:  
  
1 row(s) inserted.
```

## How it works

- If the party size is within capacity → shows an informational message.
- If the party size exceeds capacity → shows a warning message: “Our cafe follows the laws of physics – choose a larger table”
- The trigger prevents incorrect reservations and maintains realistic seating logic.

***Helps avoid double-booking or impossible reservations.***

***Improves accuracy and reliability of the reservation system.***



# Employee shift analysis(pkg\_employee\_shifts)



PROCEDURE

pr\_employee\_shift\_summary\_coll

## What this package does

- Analyzes employee work shifts over a selected period.
- Shows how many hours each employee worked.
- Measures how many hours they were late compared to planned start time.
- Finds the most late employee in the period.



```
32 | Celestia Greeves | Delivery Driver | hours=138.25 | late_h=0
3 | Krystyna Ker | Waiter | hours=138 | late_h=.58
4 | Gaby Seedhouse | Waiter | hours=136.25 | late_h=1.08
6 | Dell Hutchin | Waiter | hours=128.83 | late_h=1.33
11 | Daisy Ackeroyd | Chef | hours=123.67 | late_h=0
27 | Bjorn Feighney | Supply Controller | hours=123.25 | late_h=0
29 | El Yeld | HR Coordinator | hours=104.25 | late_h=0
```

-----  
TOTAL HOURS: 5654.5

TOTAL LATE (hours): 140.75  
-----

Most late employee: 34 - Adiana Petken | role=Dishwasher | late=40.75 hours

## PROCEDURE pr\_waste\_per\_employee;

Question: What is the total **cost of waste of each employee** ( $\text{quantity\_wasted} \times \text{Ingredients.unit\_price}$ )?

## PROCEDURE pr\_top\_waste\_reasons

Question: Group Wastage\_Log by reason\_type (or reason\_id) and sum quantity\_wasted / cost.

## PROCEDURE pr\_waste\_rate\_per\_ingredient

Question: For each ingredient, waste as a percentage of total purchased or used.

## PROCEDURE pr\_late\_delivery\_waste\_impact

Question: Is there a correlation between suppliers that often deliver late and higher waste for their ingredients?

## FUNCTION fn\_get\_daily\_waste\_cost & fn\_get\_monthly\_waste\_cost

Question: For each day/month, what is the total **cost of waste** ( $\text{quantity\_wasted} \times \text{Ingredients.unit\_price}$ )?

# Package “Waste Analysis,,



Waste per Employee

| ID | Name             | Role           | Waste Qty | Waste Cost |
|----|------------------|----------------|-----------|------------|
| 1  | Eustace Coxwell  | Waiter         | 12        | 48         |
| 11 | Daisie Ackeroyd  | Chef           | 4.87      | 29.22      |
| 16 | Loleta Esilmon   | Cook           | 4.55      | 27.3       |
| 31 | Arther Crepin    | Cashier        | 3.66      | 18.3       |
| 14 | Astra Dannel     | Cook           | 1.48      | 17.76      |
| 17 | Ted Barukh       | Cook           | 9         | 16.2       |
| 34 | Adiana Petken    | Dishwasher     | 4.55      | 8.19       |
| 7  | Hansiain Sibbons | Waiter         | 4.97      | 5.96       |
| 22 | Solly Arthan     | Barista        | 2.63      | 4.73       |
| 5  | Abby Hesey       | Waiter         | .95       | 3.61       |
| 4  | Gaby Seedhouse   | Waiter         | .68       | 2.58       |
| 33 | Nedda Eneas      | Dishwasher     | 1.36      | 2.45       |
| 24 | Joaquin Emmett   | Cleaner        | 1.34      | 1.21       |
| 29 | El Yeld          | HR Coordinator | .26       | 1.17       |
| 18 | Odella Tabary    | Cook           | .9        | .09        |

Results:



Top Waste Reasons

|                                   |                        |              |
|-----------------------------------|------------------------|--------------|
| Reason: Expired                   | Quantity wasted: 10.48 | Cost: 33.96  |
| Reason: Spilled                   | Quantity wasted: 7.5   | Cost: 33.954 |
| Reason: Supplier quality issue    | Quantity wasted: 9.52  | Cost: 33.264 |
| Reason: Damaged during storage    | Quantity wasted: 3.92  | Cost: 19.47  |
| Reason: Over-prepared             | Quantity wasted: 6.57  | Cost: 11.98  |
| Reason: Wrong temperature storage | Quantity wasted: .95   | Cost: 3.61   |
| Reason: Burned during cooking     | Quantity wasted: 2.26  | Cost: 2.538  |
| Reason: Expiry                    | Quantity wasted: -2    | Cost: -8     |
| Reason: Damage                    | Quantity wasted: -4    | Cost: -16    |
| Reason: Overproduction            | Quantity wasted: -6    | Cost: -24    |





# Results:

| Waste Rate per Ingredient |                    |           |              |         |
|---------------------------|--------------------|-----------|--------------|---------|
| ID                        | Name               | Waste Qty | Consumed Qty | Waste % |
| 10                        | Vanilla Syrup      | 12.15     | 13.76        | 88.3    |
| 78                        | Spinach            | 17.3      | 22.22        | 77.86   |
| 74                        | Syrup              | 22.21     | 38.13        | 58.25   |
| 114                       | Lemon Juice        | 9.14      | 17.09        | 53.48   |
| 40                        | Butter             | 2.63      | 5.87         | 44.8    |
| 53                        | Almonds            | 8.99      | 22.39        | 40.15   |
| 67                        | Tomatoes           | 9.86      | 25.53        | 38.62   |
| 11                        | Caramel Syrup      | 1.39      | 3.64         | 38.19   |
| 82                        | Bread              | 19.02     | 50.25        | 37.85   |
| 26                        | Jasmine Tea Leaves | 4.5       | 11.96        | 37.63   |
| 39                        | Baking Soda        | 3.25      | 9.1          | 35.71   |
| 56                        | Breadcrumbs        | 4.06      | 11.62        | 34.94   |
| 18                        | Honey              | 1.48      | 4.3          | 34.42   |
| 106                       | Cucumbers          | 5.69      | 16.67        | 34.13   |
| 105                       | Cherry Tomatoes    | 4.55      | 13.34        | 34.11   |
| 44                        | Heavy Cream        | 1.31      | 3.86         | 33.94   |
| 16                        | Sugar              | 1.51      | 4.47         | 33.78   |
| 94                        | Carrots            | 16        | 50           | 32      |
| 13                        | Chocolate Syrup    | 1.63      | 5.13         | 31.77   |
| 63                        | Tortilla Chips     | 5.71      | 18.07        | 31.6    |
| 38                        | Baking Powder      | 3.4       | 10.82        | 31.42   |
| 46                        | Blueberries        | 6.58      | 22.23        | 29.6    |
| 84                        | Jam                | 6.5       | 23.64        | 27.5    |
| 52                        | Yeast              | 7.49      | 28.73        | 26.07   |
| 1                         | Coffee Beans       | .73       | 2.81         | 25.98   |
| 88                        | Shrimp             | 10        | 41           | 24.39   |

## Impact of Late Deliveries on Waste

|                                |                        |                   |                    |
|--------------------------------|------------------------|-------------------|--------------------|
| Supplier: Patin Egarr          | Late deliveries: 7/7   | Late rate: 100%   | Total waste: 5.81  |
| Supplier: Donnajean Shambrook  | Late deliveries: 7/7   | Late rate: 100%   | Total waste: 4.18  |
| Supplier: Larine Edgley        | Late deliveries: 9/10  | Late rate: 90%    | Total waste: 4.18  |
| Supplier: Arlena Akid          | Late deliveries: 6/7   | Late rate: 85.71% | Total waste: 10.59 |
| Supplier: Melony Gerwood       | Late deliveries: 18/21 | Late rate: 85.71% | Total waste: 20.29 |
| Supplier: Richy Brookesbie     | Late deliveries: 6/7   | Late rate: 85.71% | Total waste: 26.55 |
| Supplier: Jarrett McNaught     | Late deliveries: 17/21 | Late rate: 80.95% | Total waste: 15.71 |
| Supplier: Vincent Broszkiewicz | Late deliveries: 17/21 | Late rate: 80.95% | Total waste: 39.64 |
| Supplier: Henderson Yeskin     | Late deliveries: 14/18 | Late rate: 77.78% | Total waste: 1.04  |
| Supplier: Monah Kubasiewicz    | Late deliveries: 10/14 | Late rate: 71.43% | Total waste: 14.27 |
| Supplier: Bone Dahler          | Late deliveries: 5/7   | Late rate: 71.43% | Total waste: 18.89 |
| Supplier: Hermon Mansbridge    | Late deliveries: 18/28 | Late rate: 64.29% | Total waste: 9.35  |
| Supplier: Antoinette Seth      | Late deliveries: 8/14  | Late rate: 57.14% | Total waste: 24.92 |
| Supplier: Alonzo Pogue         | Late deliveries: 4/7   | Late rate: 57.14% | Total waste: 5.35  |
| Supplier: Susi Van Velde       | Late deliveries: 4/7   | Late rate: 57.14% | Total waste: 4.18  |
| Supplier: Renae Poulton        | Late deliveries: 11/21 | Late rate: 52.38% | Total waste: 25.61 |
| Supplier: Emery Grishmanov     | Late deliveries: 13/28 | Late rate: 46.43% | Total waste: 30.12 |
| Supplier: Gideon Stayte        | Late deliveries: 6/14  | Late rate: 42.86% | Total waste: 7.4   |
| Supplier: Lianne Tew           | Late deliveries: 6/14  | Late rate: 42.86% | Total waste: 27.01 |
| Supplier: Dinny Mumford        | Late deliveries: 5/14  | Late rate: 35.71% | Total waste: 10.58 |
| Supplier: Melina Gowen         | Late deliveries: 10/28 | Late rate: 35.71% | Total waste: 23.3  |
| Supplier: Maiga Lowdyane       | Late deliveries: 7/21  | Late rate: 33.33% | Total waste: 19.96 |
| Supplier: Trixi Coggins        | Late deliveries: 9/28  | Late rate: 32.14% | Total waste: 27.39 |
| Supplier: Randi Ackerley       | Late deliveries: 9/28  | Late rate: 32.14% | Total waste: 35.42 |
| Supplier: Greer Morgans        | Late deliveries: 6/21  | Late rate: 28.57% | Total waste: 36.72 |
| Supplier: Omar Quene           | Late deliveries: 10/35 | Late rate: 28.57% | Total waste: 20.07 |

### WASTE\_COST\_FOR\_12\_NOV

26.49

1 rows returned in 0.01 seconds [Download](#)

### NOV\_WASTE\_COST

122.576

1 rows returned in 0.02 seconds [Download](#)



# Triggers

## 1) `trg_wastage_reason_require`

This trigger ensures data integrity by blocking any INSERT or UPDATE into `wastage_log` if the field `reason_type` is missing or empty. It guarantees that every recorded waste event has a clear, mandatory reason, preventing incomplete or unclear waste entries

## 2) `trg_expiring_ingredient_alert`

This alert mechanism tracks ingredients that are close to their expiry date and writes human-readable warnings into the `INGREDIENT_ALERTS` table. The trigger automatically creates an `EXPIRING_SOON` alert when a new delivery with near expiry is inserted, and the procedure (like `pr_fill_expiry_alerts`) can be run to rescan all supply orders, delete old alerts, and regenerate up-to-date expiry warnings



## 1) PROCEDURE pr\_revenue\_by\_payment\_method;

*Question: What share of total revenue is paid by cash, card, QR, etc.?*

*Use: Decide which payment methods to promote/keep; optimize fees.*

## 2) PROCEDURE pr\_revenue\_by\_payment\_method;

*Question: How much revenue is lost due to  
**canceled orders** (order\_status = Canceled) vs  
total revenue?*

*Use: See how critical cancellations are; maybe  
create deposit policy for reservations.*

# Package Financial Analysis



# Results:

| Results                   | Explain | Describe | Saved SQL | History |
|---------------------------|---------|----------|-----------|---------|
| Revenue by Payment Method |         |          |           |         |
| -----                     |         |          |           |         |
| Method                    | Revenue | Percent  |           |         |
| Online                    | 4217    | 35.42%   |           |         |
| Cash                      | 3777    | 31.72%   |           |         |
| Card                      | 3700    | 31.08%   |           |         |



| Results                       | Explain | Describe | Saved SQL | History |
|-------------------------------|---------|----------|-----------|---------|
| Refunds / Cancellation Impact |         |          |           |         |
| -----                         |         |          |           |         |
| Revenue (Ready Orders):       |         |          | 9530      |         |
| Lost Revenue (Cancelled):     |         |          | 2376      |         |
| Percent Lost:                 |         |          | 19.96%    |         |

**Thank You  
Very Much!**

